

Mehmet Mert Kutlukaya

Mersin, Türkiye · +90 546 850 3304 · mmkutlukaya@gmail.com · linkedin.com/in/mehmet-mert-kutlukaya · github.com/HolyRock-yottabyte

SUMMARY

Software developer and CTIS student with hands-on experience in embedded systems, autonomous UAV software, and robotic control systems. TEKNOFEST International UAV Competition finalist. Skilled in C/C++, sensor integration, PID control, and real-time image processing — seeking to apply these skills in a mechatronics engineering environment.

PROJECTS

AeroForge DI - Physics-Based Drone Engineering SaaS Platform

Jul 2026 - Present

- Built a full-stack SaaS platform (Python/FastAPI + vanilla JS) that unifies motor, propeller, ESC, and battery selection into a single physics-based analysis engine with 12 engineering models (momentum theory, Peukert battery model, thermal/structural analysis).
- Engineered a 236+ component library with real manufacturer data and thrust curves; implemented AI-powered optimization that scans thousands of configurations to maximize flight time and minimize hardware cost.
- Designed a responsive dark-theme UI with real-time result rendering, interactive throttle curves, PDF/ArduPilot .param export, multi-language support (EN/TR), and a role-based admin panel with usage analytics.

Autonomous Drone - TEKNOFEST International UAV Competition (Finalist)

Mar - Sep 2025

- Developed autonomous flight software on Raspberry Pi 5: sensor fusion, mission planning, Pixhawk/ArduPilot integration with PID stabilization, GPS positioning, and telemetry logging (TEKNOFEST 2025 Finalist).
- Implemented OpenCV target detection (rectangle/hexagon) for autonomous orientation, tracking, and precision landing; extended with obstacle detection/avoidance and Wi-Fi API remote control (Python/Flask) for 2026 iteration.
- Built and refined full autonomous UAV pipeline from takeoff to precision landing; currently enhancing with advanced computer vision tracking for TEKNOFEST 2026 competition.

Mars Rover - Robotic Arm Control

Sep 2025 - Present

- STM32 (C/C++) PID joint control and inverse kinematics for precise soil-sampling arm operations.
- Real-time image processing for soil/target analysis; STM32-mission computer comms via Serial/CAN bus.

PetCare Veterinary Mobile App

May 2025 - Dec 2025

- Cross-platform Flutter/Dart app with Bloc state management, clean architecture, and cloud backend integration.

Visual Vibrometry & ML-Based Predictive Maintenance - TUBITAK 2242

24 Apr 2026 - Present

- Developed a contactless predictive maintenance pipeline to classify mechanical faults in rotating equipment using video-based vibration analysis.
- Implemented object tracking and power spectral density (PSD) analysis with harmonic/order-based feature extraction in MATLAB.
- Evaluated multiple tracking algorithms and ML models; achieved 86.7% blind holdout accuracy across healthy, unbalance, misalignment, and mechanical looseness fault classes.
- Ongoing research project; physical test-rig validation and expanded fault classification planned as the next development stage.

WORK EXPERIENCE

Intern | GMK Air Aviation Technologies

Jun 2026 - Sep 2026 · Mersin

- Worked on UAV system integration, embedded hardware, sensor interfaces, telemetry, power systems, and flight-control configurations across multiple development and test stages.
- Performed hardware/software testing, calibration, system diagnostics, and troubleshooting for communication, navigation, control, and flight-stability issues.
- Supported simulation, ground testing, and real-flight validation; analyzed telemetry and flight data to identify system behavior and improve reliability.
- Contributed to control-system and autonomous-flight studies, including PID tuning, sensor integration, flight modes, and mission-oriented UAV workflows.
- Prepared wiring diagrams, integration notes, test reports, troubleshooting documentation, and technical user manuals for engineering and operational use.

Game Developer | EARTUP

Sep 2022 - Feb 2024 · Istanbul

- Designed and implemented hypercasual game mechanics in Unity (C#) for mobile platforms.
- Optimized game performance and memory usage to ensure smooth gameplay across target devices.
- Led QA processes including test case design, bug tracking, and regression testing workflows.

QA / Game Tester | Vanquisher Games

Feb - Jun 2022 · Istanbul

- Performed functional and performance testing on pre-release game builds across multiple platforms.
- Authored detailed bug reports and maintained comprehensive test documentation for the development team.
- Conducted fix validation and regression testing to ensure software quality and release readiness.

EDUCATION

B.Sc. Information Systems & Technologies (CTIS)

2023 - Present

Mersin University · Mersin, Türkiye

LANGUAGES

Turkish · Native | English · B2 (Upper-Intermediate)

TECHNICAL SKILLS

Embedded & Hardware: STM32, Raspberry Pi, Pixhawk/ArduPilot, MPU6050, GPS, barometer, Serial/CAN

Control Systems: PID control, inverse kinematics, sensor fusion, autonomous navigation

Languages: C, C++, Python, C#, Dart, JavaScript, R

Computer Vision: OpenCV, YOLOv11, TensorFlow/Keras, real-time pipelines

Mobile: Flutter/Dart, Bloc, clean architecture, REST API

Other: Git/GitHub, Docker, Linux (Kali/Ubuntu), Unity, Unreal Engine, FastAPI, ASP.NET

COMMUNITY & PROGRAMS

Organizer Member · Google Developer Groups (GDG) — Present

Participant (Program In Progress) · SiberVatan Cybersecurity Training Program — Present

AWARDS

- TEKNOFEST International UAV Competition — Finalist · TEKNOFEST, Sep 2025
- SiberVatan 2026 Program — 3rd Place · SiberVatan, 2026
- TEKNOFEST 2026 NSosyal Innovation Competition Finalist · TEKNOFEST, Sep 2026

CERTIFICATIONS

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|---|---|
| • Enterprise Security in Practice · IBM, Jun 2026 | • Kali Linux · Board Infinity / Coursera, Oct 2025 |
| • Getting Started with Threat Intelligence and Hunting · IBM, Jun 2026 | • TEKNOFEST 2025 International UAV Competition Final Participation Document, Sep 2025 |
| • Governance, Risk, Compliance, and Data Privacy · IBM, Jun 2026 | • Introduction to Large Language Models · BTK Akademi, Mar 2026 |
| • Turkey Cyber Vatan Program 2026 Participation Certificate · SiberVatan, Jun 2026 | • Intro to Back-End Development · Meta / Coursera, Nov 2025 |
| • Turkey Cyber Vatan Program 2026 Final Participation Document · SiberVatan, Jun 24, 2026 | • Intro to Front-End Development · Meta / Coursera, Nov 2025 |
| • Certified Associate Penetration Tester (CAPT) · Hackviser, Feb 2026 | • Network Security · Cisco, Jun 2022 |
| • Computer Vision with YOLOv11 · BTK Akademi, Mar 2026 | • Introduction to Cybersecurity · Cisco, Jul 2022 |
| • Python & TensorFlow for Data Science · BTK Akademi, Mar 2026 | • IBM x Kodluyoruz: CyberStart 2.0 - Advanced Level · Kodluyoruz, Jul 2026 |
| • IBM x Kodluyoruz: AI4Future - Advanced Level · Kodluyoruz, Jul 2026 | • Make Agentic AI Work for You · IBM, Jul 2026 |
| • Build Smarter AI with Embeddings · IBM, Jul 2026 | |